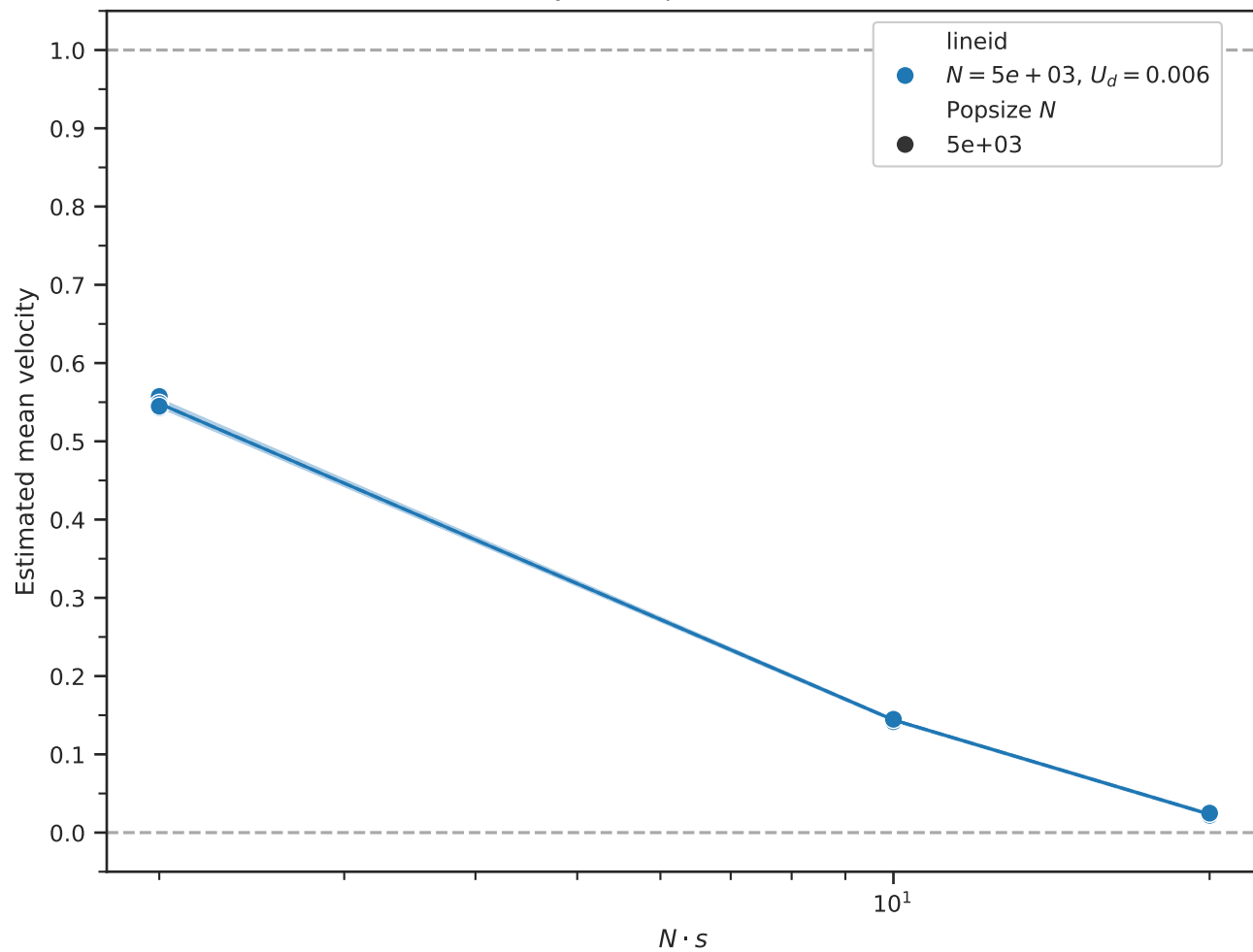
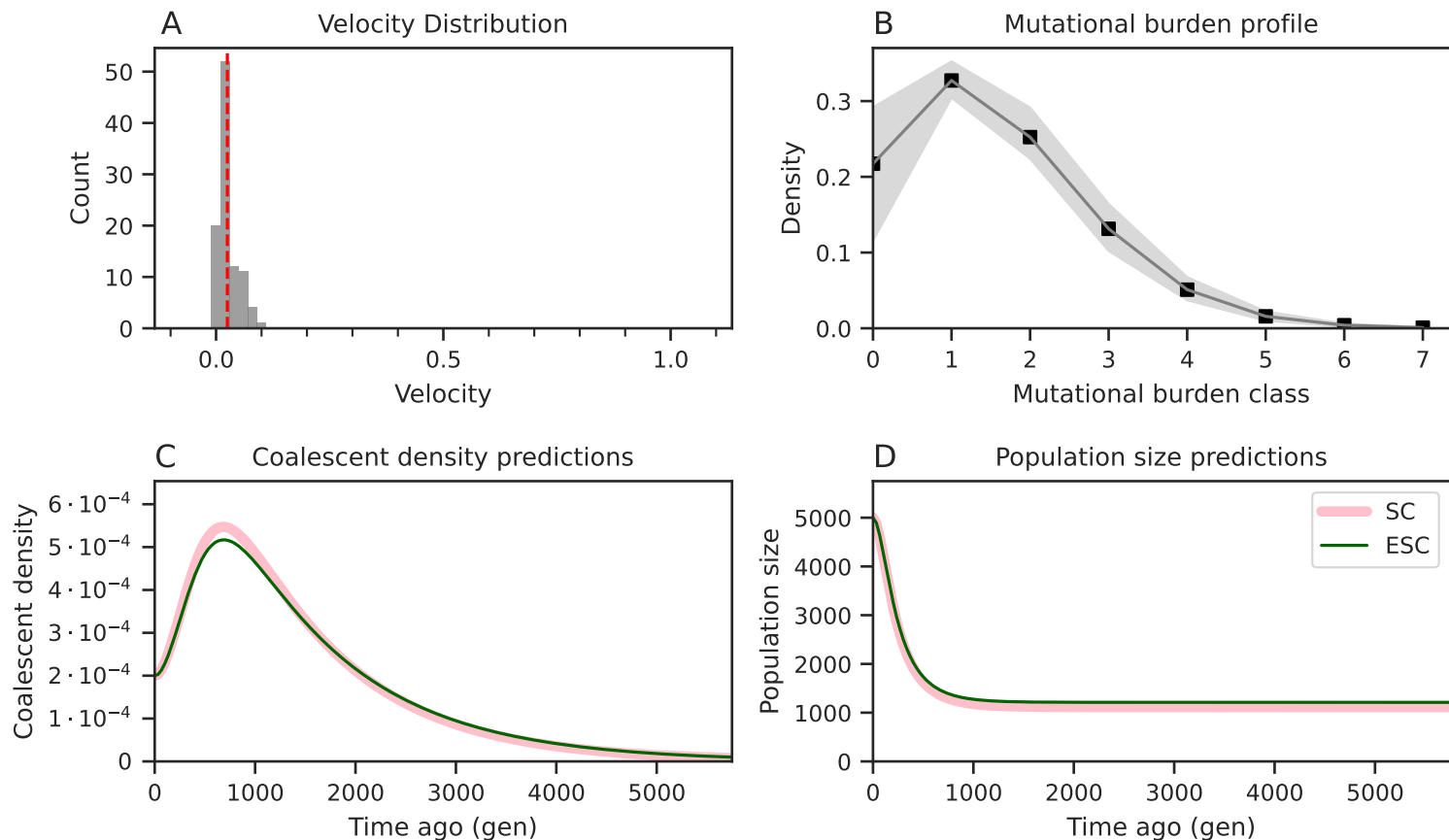


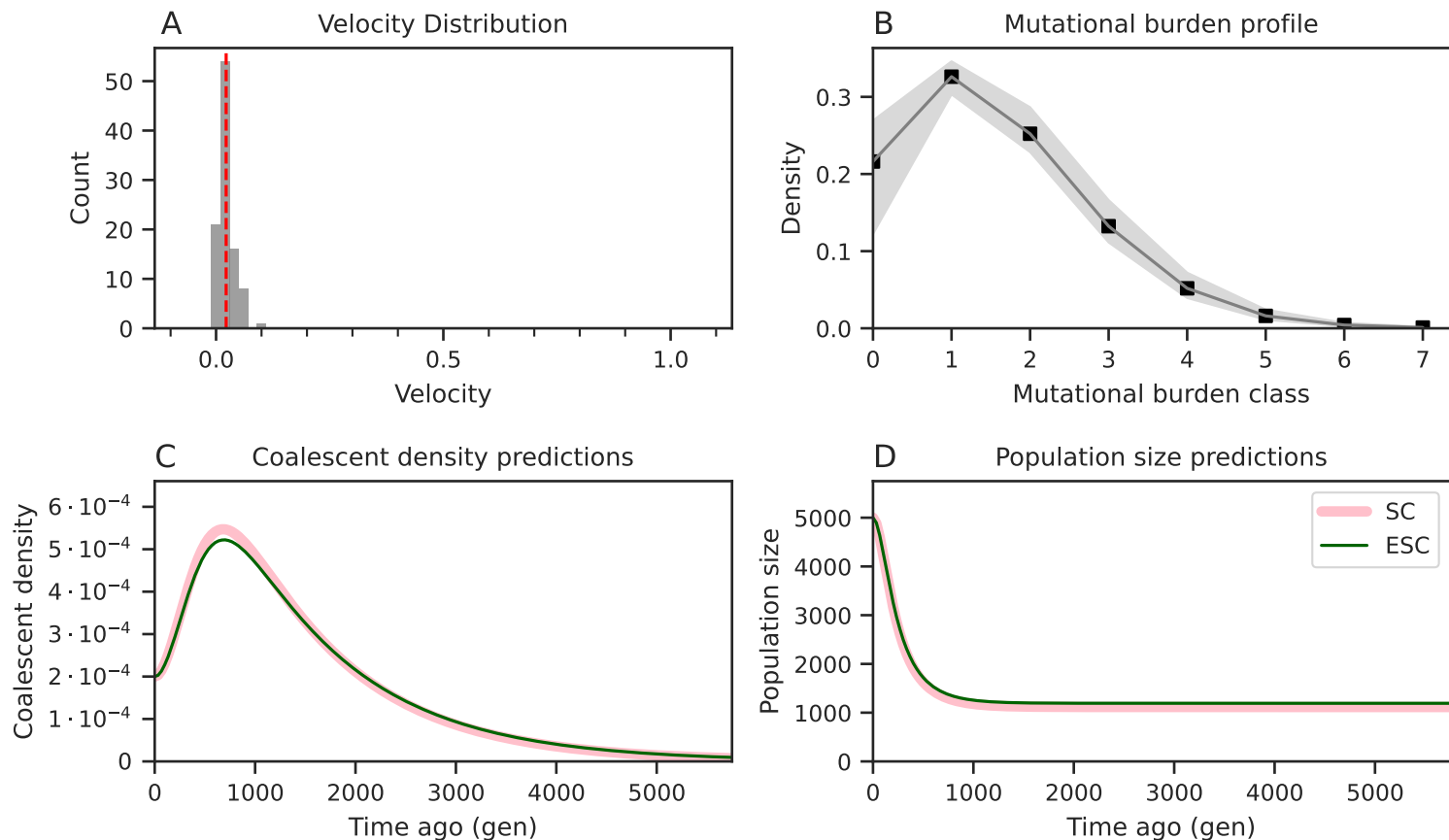
Mean velocity across parameter combinations



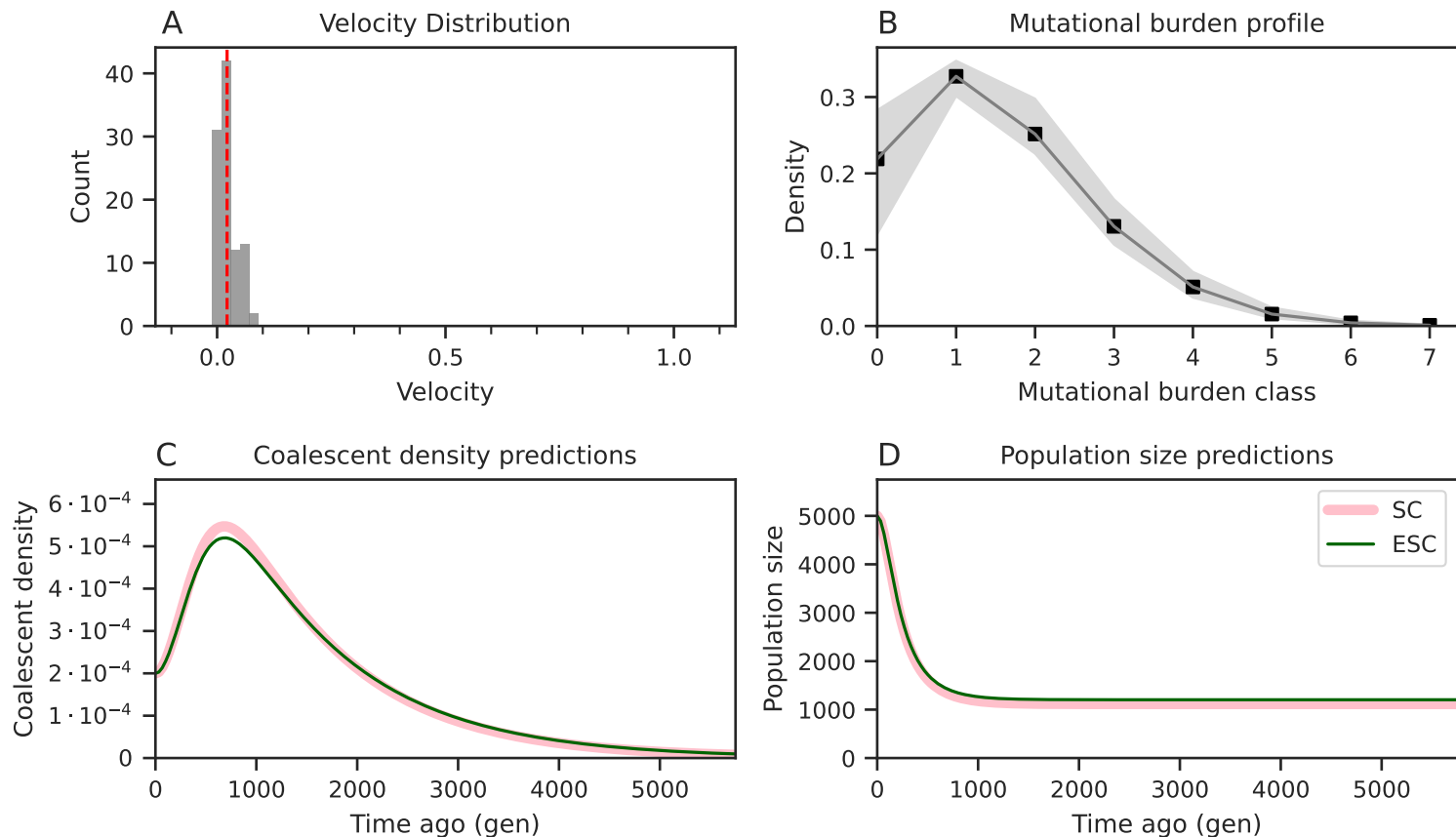
$N = 5e + 03$, $s = 0.004$, $U_d = 0.006$, $\sigma = 0.002$, $\phi = 4.46$
Mean Velocity= 0.0249 ± 0.0208 , $n = 100$ simulations



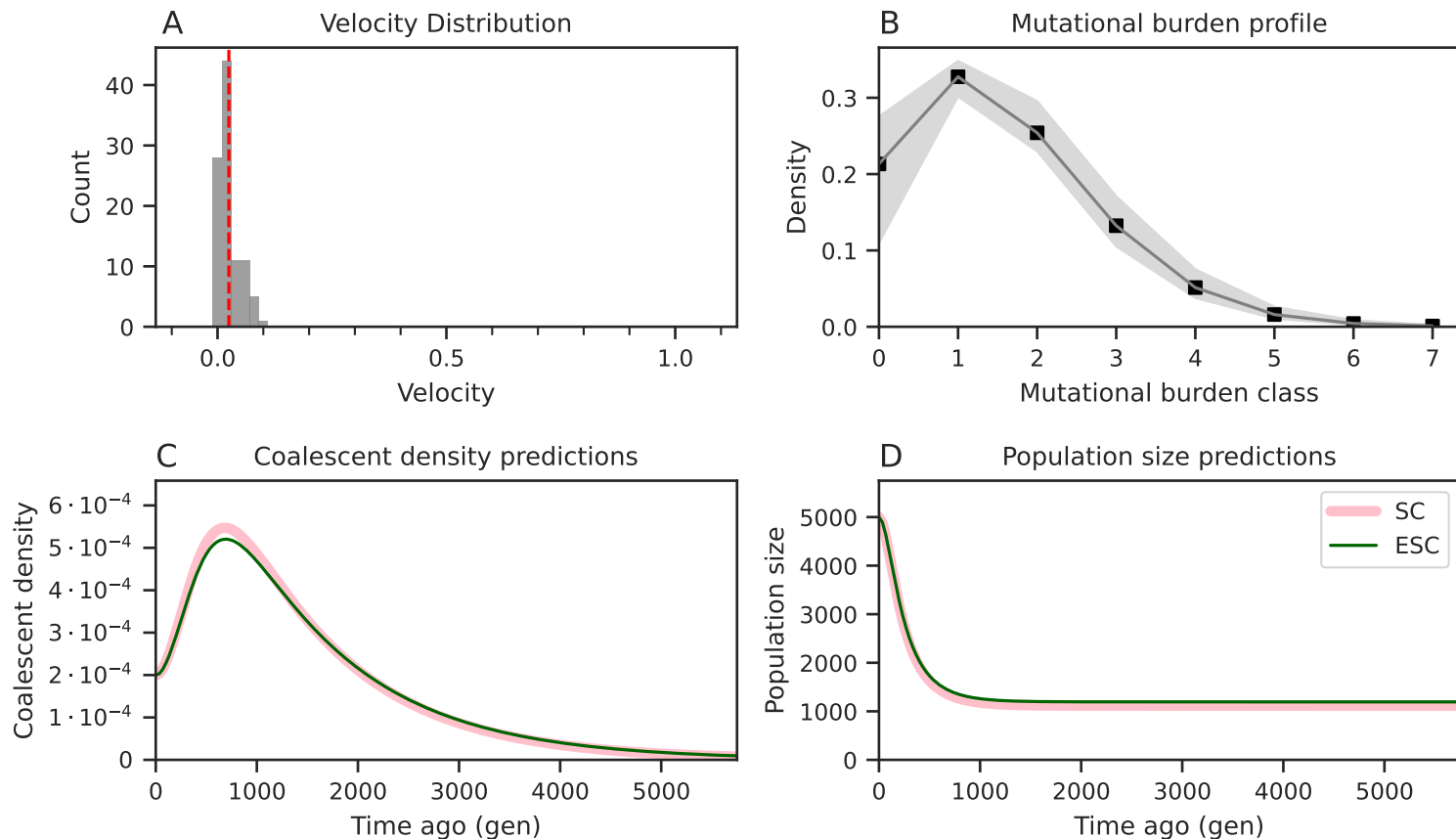
$N = 5e + 03$, $s = 0.004$, $U_d = 0.006$, $\sigma = 0.001$, $\phi = 4.46$
Mean Velocity= 0.0222 ± 0.0168 , $n = 100$ simulations



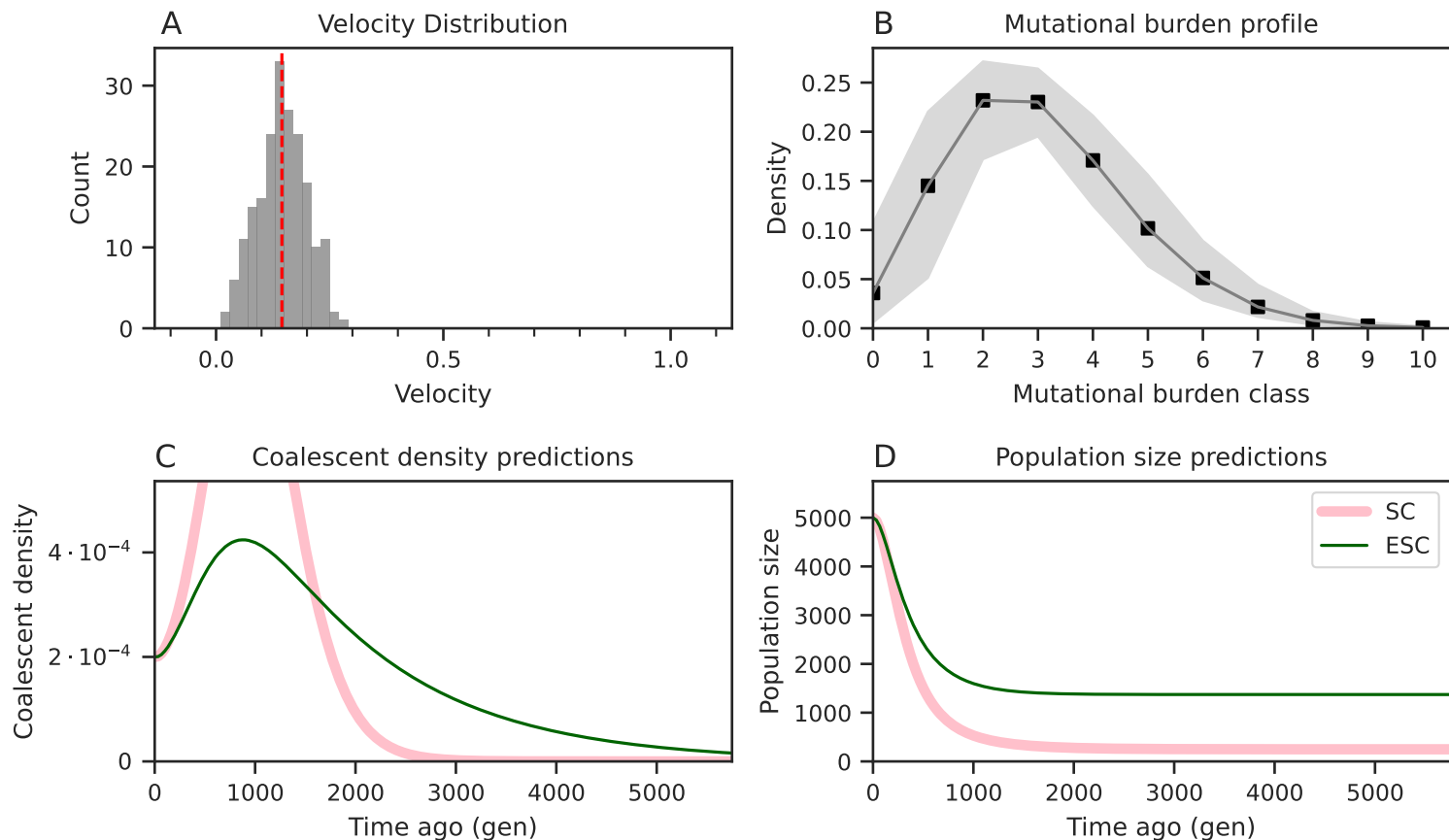
$N = 5e + 03$, $s = 0.004$, $U_d = 0.006$, $\sigma = 5 \cdot 10^{-4}$, $\phi = 4.46$
Mean Velocity= 0.0219 ± 0.0195 , $n = 100$ simulations



$N = 5e + 03$, $s = 0.004$, $U_d = 0.006$, $\sigma = 3 \cdot 10^{-4}$, $\phi = 4.46$
Mean Velocity= 0.0247 ± 0.0217 , $n = 100$ simulations

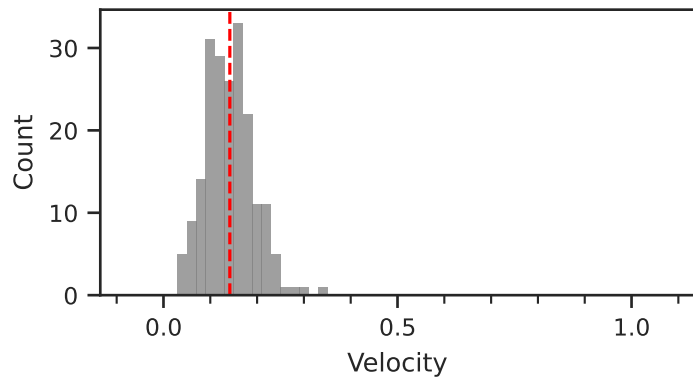


$N = 5e + 03$, $s = 0.002$, $U_d = 0.006$, $\sigma = 0.001$, $\phi = 0.498$
Mean Velocity= 0.1448 ± 0.0544 , $n = 200$ simulations

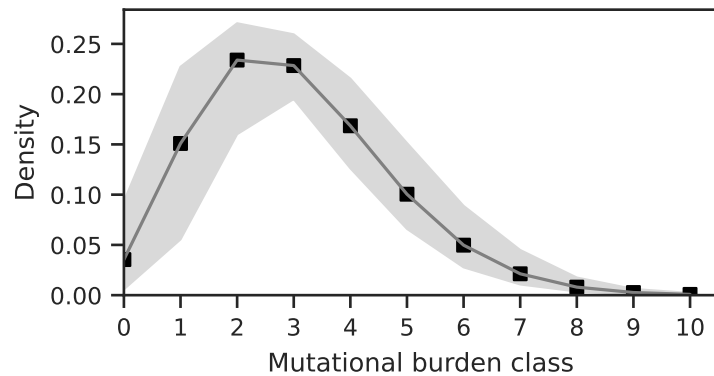


$N = 5e + 03$, $s = 0.002$, $U_d = 0.006$, $\sigma = 5 \cdot 10^{-4}$, $\phi = 0.498$
Mean Velocity= 0.1418 ± 0.0506 , $n = 200$ simulations

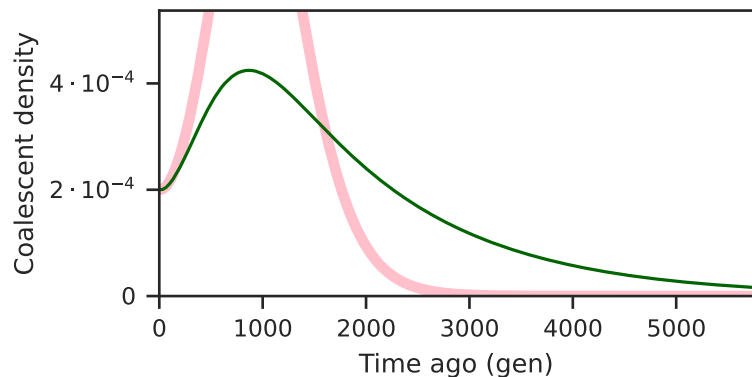
A Velocity Distribution



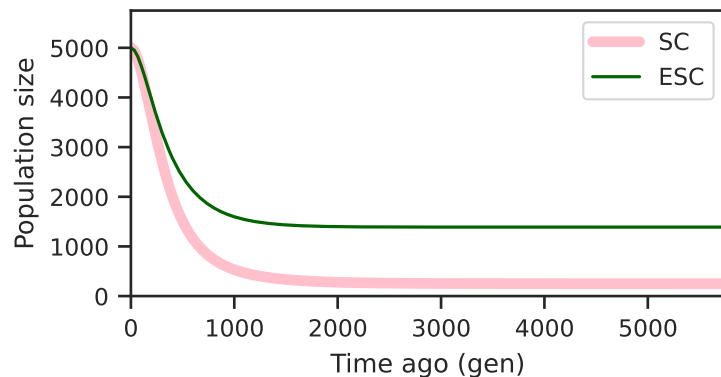
B Mutational burden profile



C Coalescent density predictions

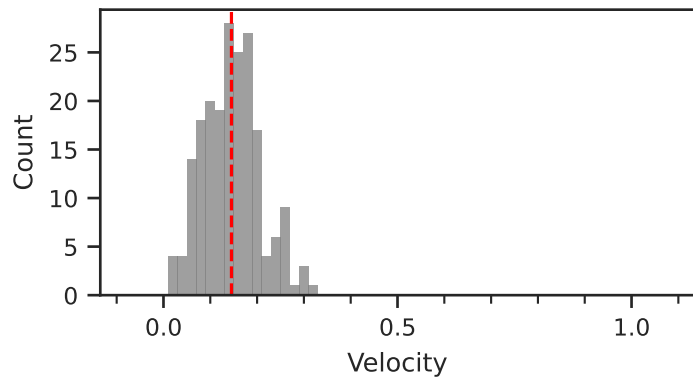


D Population size predictions

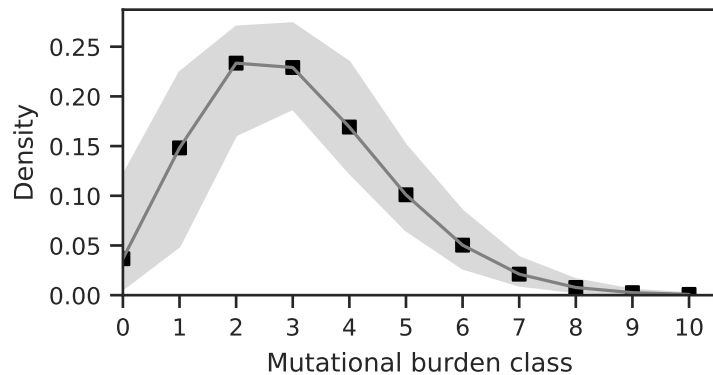


$N = 5e + 03$, $s = 0.002$, $U_d = 0.006$, $\sigma = 3 \cdot 10^{-4}$, $\phi = 0.498$
Mean Velocity= 0.1452 ± 0.0600 , $n = 200$ simulations

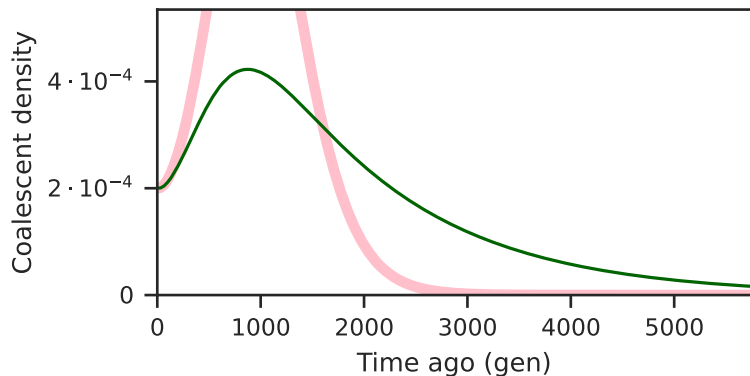
A Velocity Distribution



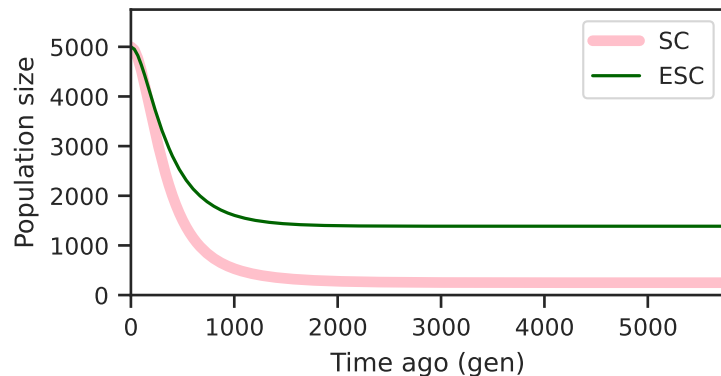
B Mutational burden profile



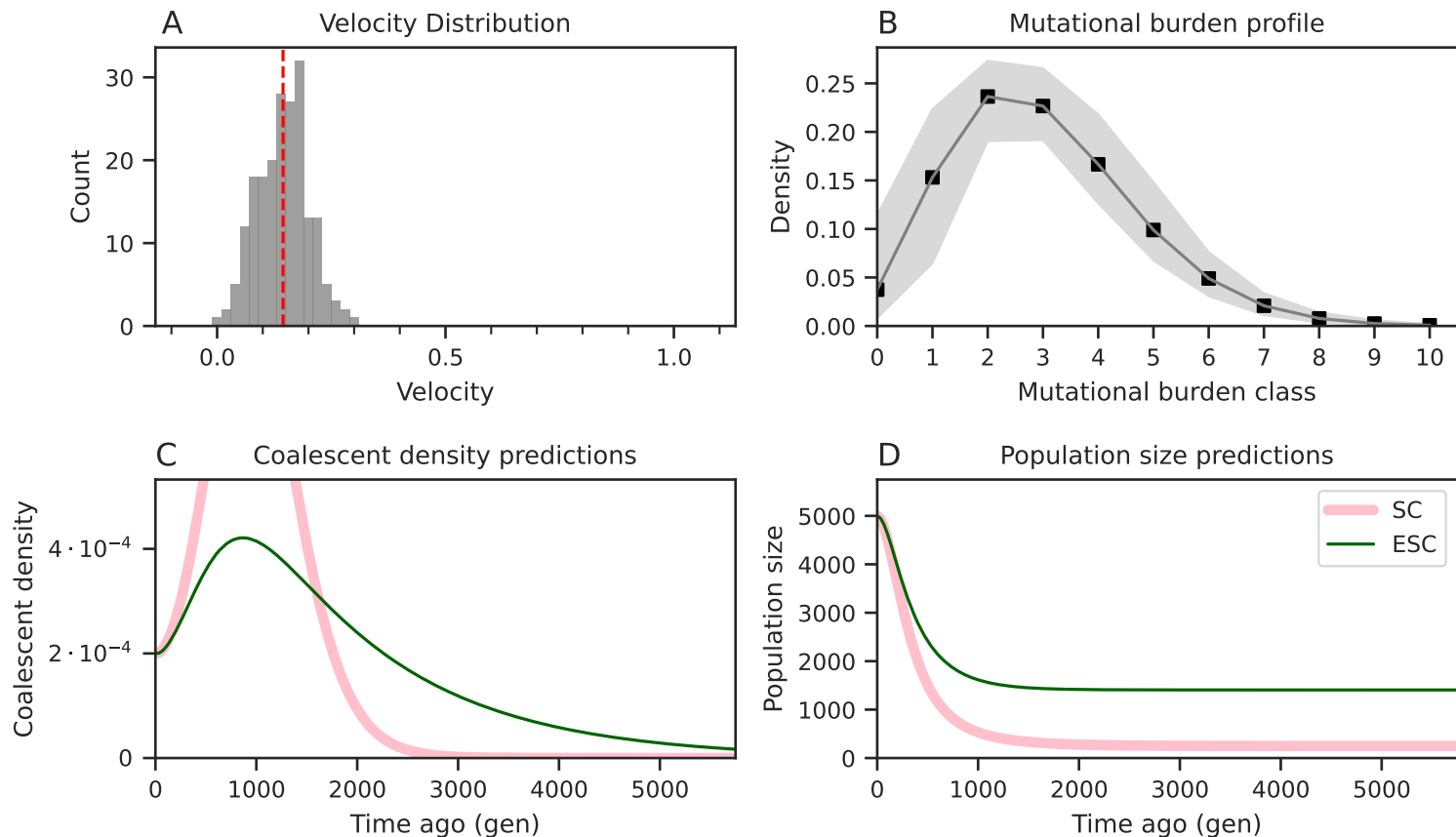
C Coalescent density predictions



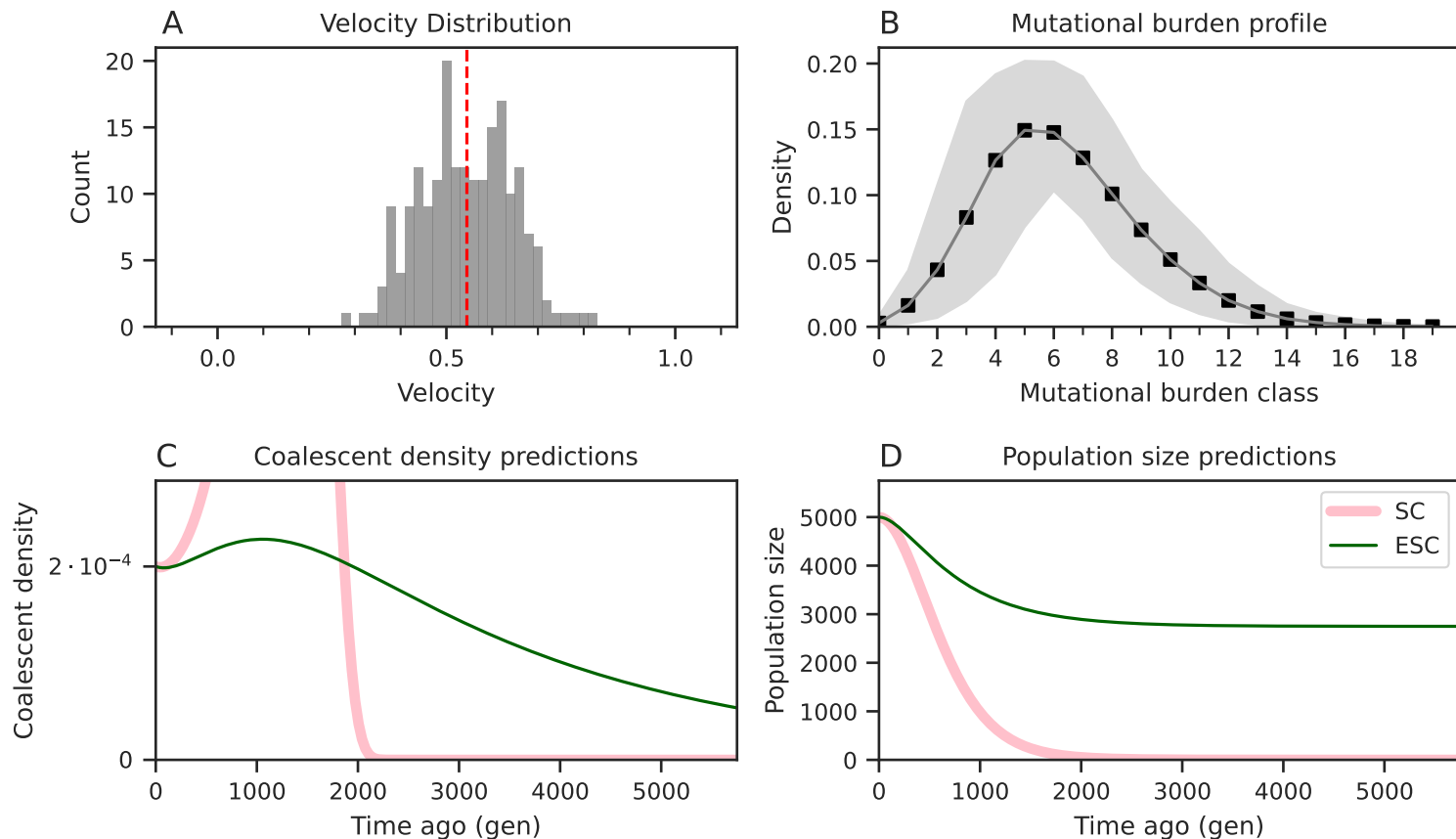
D Population size predictions



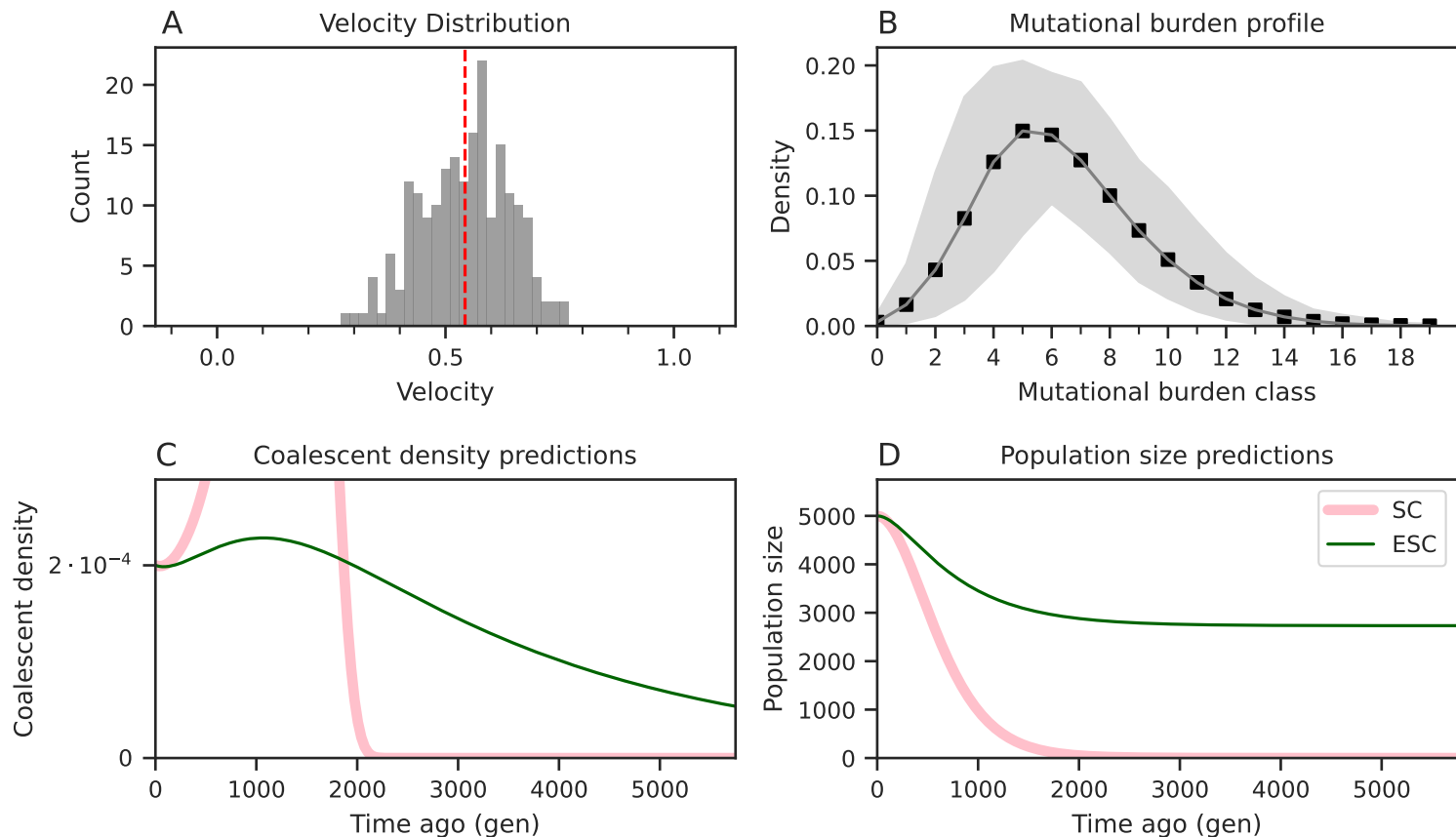
$N = 5e + 03$, $s = 0.002$, $U_d = 0.006$, $\sigma = 2 \cdot 10^{-4}$, $\phi = 0.498$
Mean Velocity = 0.1443 ± 0.0559 , $n = 200$ simulations



$N = 5e + 03$, $s = 4 \cdot 10^{-4}$, $U_d = 0.006$, $\sigma = 2 \cdot 10^{-4}$, $\phi = 6.12 \cdot 10^{-7}$
Mean Velocity = 0.5448 ± 0.1004 , $n = 200$ simulations

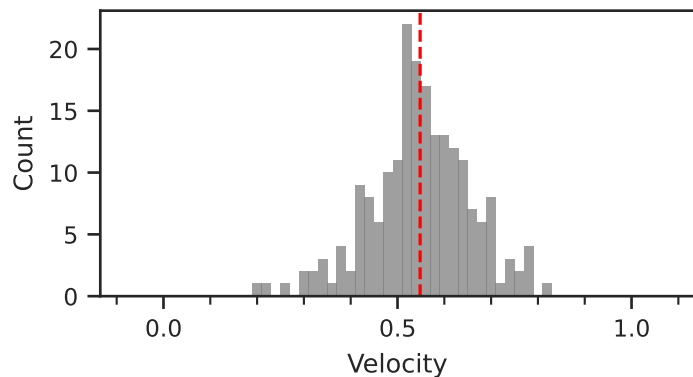


$N = 5e + 03$, $s = 4 \cdot 10^{-4}$, $U_d = 0.006$, $\sigma = 1 \cdot 10^{-4}$, $\phi = 6.12 \cdot 10^{-7}$
Mean Velocity = 0.5428 ± 0.0980 , $n = 200$ simulations

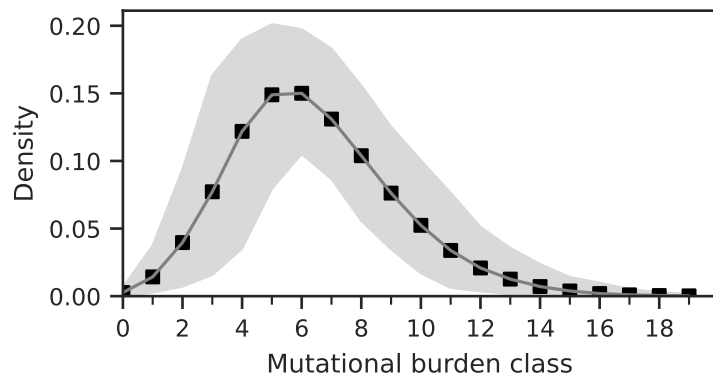


$N = 5e + 03$, $s = 4 \cdot 10^{-4}$, $U_d = 0.006$, $\sigma = 5 \cdot 10^{-5}$, $\phi = 6.12 \cdot 10^{-7}$
Mean Velocity = 0.5483 ± 0.1074 , $n = 200$ simulations

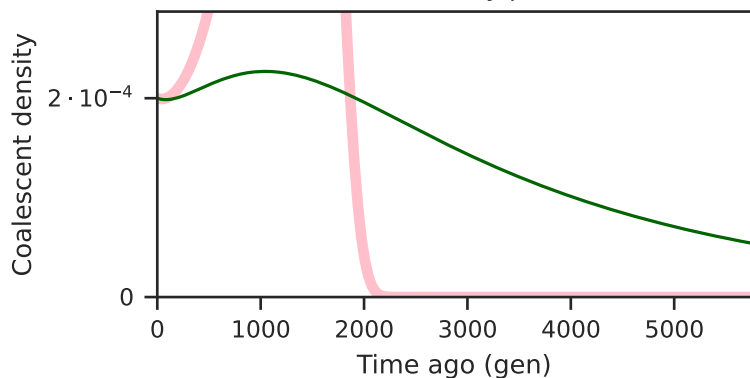
A Velocity Distribution



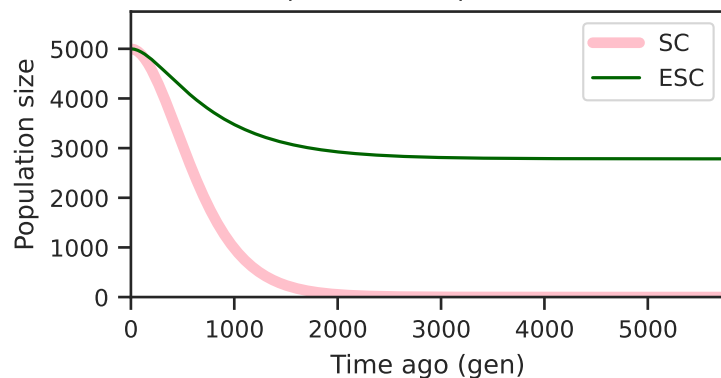
B Mutational burden profile



C Coalescent density predictions

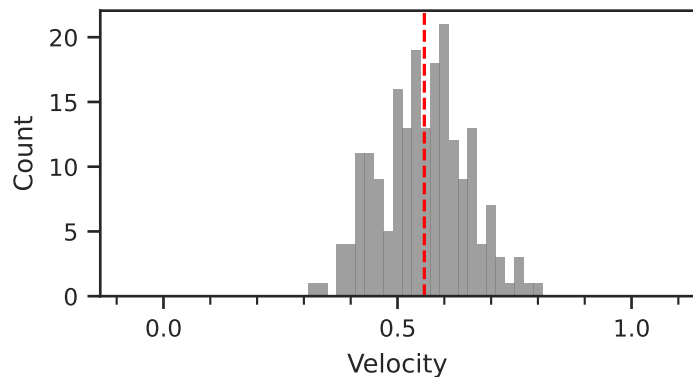


D Population size predictions

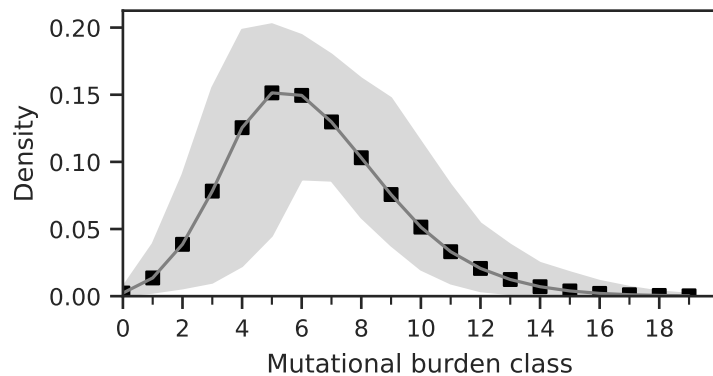


$N = 5e + 03$, $s = 4 \cdot 10^{-4}$, $U_d = 0.006$, $\sigma = 3 \cdot 10^{-5}$, $\phi = 6.12 \cdot 10^{-7}$
Mean Velocity = 0.5573 ± 0.0916 , $n = 200$ simulations

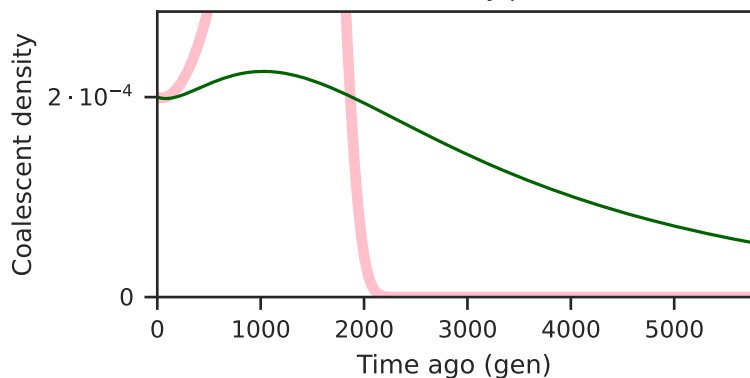
A Velocity Distribution



B Mutational burden profile



C Coalescent density predictions



D Population size predictions

